input becomes, can be called an *infinite dynamic range.* This crucial property arises because the inputs *multiply* the activities, in both the on-center and off-surround of the network. This multiplicative property is also called *automatic gain control* because it enables the circuit to work well in response to a wide range of input sizes. The terms where inputs multiply activities control the equilibrium activities and also the rates with which activities converge to equilibrium, with larger inputs causing faster reaction rates.

As I noted in Figure 1.8, the property of total activity normalization also occurs in recurrent, or feedback, shunting on-center off-surround networks (Figure 1.6) that are used to store input patterns in short term memory. Total activity normalization implies a *limited capacity* of the network (Posner, 1980). Because of limited capacity, when more cells compete, each one will tend to be less active. Said in another way, increasing the activity of one cell comes at the cost of decreasing the activities of other cells. When too many inputs occur, some of their cells may thus not be activated enough to overcome their quenching threshold, and will not be stored in short term memory (Figure 1.9).

The sum of all of the activities is bounded by a constant in a non-recurrent on-center off-surround network (Figure 2.23). This sum converges through time to a constant in a recurrent network (Figure 1.7). Activities in such a recurrent network thus behave like probabilities that are dynamically changing through time. A larger activity may be interpreted to mean that the features coded by its cell population are more probable. Various of the neural architectures that this book describes may likewise be interpreted as carrying out one or another kind of probabilistic decision making, planning, or the like.

Deriving the membrane equations of neurophysiological from a thought experiment. The mass action, or shunting, dynamics that we have been led to by the thought experiment are the dynamics used by neurons. They are also called the membrane equations of neurophysiology. The equation at the top of Figure 2.24 shows this familiar equation for the variable voltage V of a cell in terms of its capacitance C; three constant saturation voltages: an excitatory saturation voltage V^+, an inhibitory saturation voltage V^-, and a passive saturation voltage V^p; and three conductances that may or may not vary through time: an excitatory conductance g^+, an inhibitory conductance g^-, and a passive conductance g^p. When these variables are set equal to the values in the lower right of Figure 2.24, the result is the equation at the top of Figure 2.23. We have hereby been led to derive the membrane equations of neurophysiology from our thought experiment.

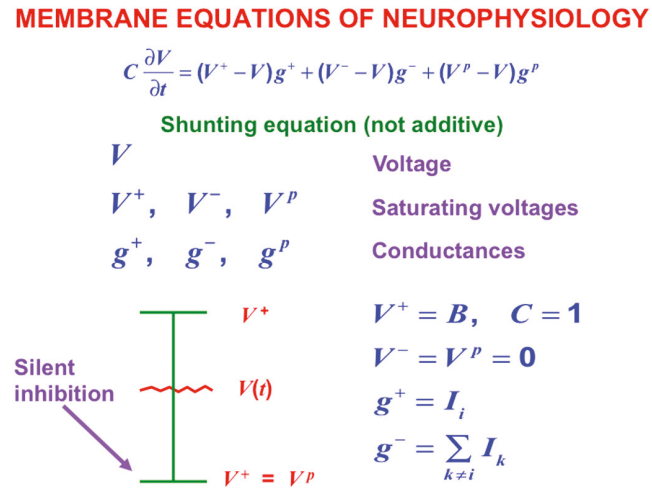

FIGURE 2.24. The membrane equations of neurophysiology describe how cell voltages change in response to excitatory, inhibitory, and passive input channels. Each channel is described by a potential difference multiplied by a conductance. With the special choices shown in the lower right hand corner, this equation defines a feedforward shunting on-center off-surround network.

Towards a universal developmental code using mass action cooperative-competitive dynamics

We have actually done a lot more than that, because this kind of shunting dynamics solves the noise-saturation dilemma *only* if it is carried out an on-center off-surround anatomy. This basic fact raises the question: Did the membrane equations and on-center off-surround anatomies *co-evolve* during the evolutionary process, so that cell networks could process distributed feature patterns without noise or saturation from the earliest stages of the evolution of cellular organisms?

Two languages can equally well describe this result, and one of them generalizes to all cellular tissues: On the one hand, there are the shunting on-center off-surround anatomies that are found in our brains. On the other hand, there are the mass action cooperative-competitive systems that can be found in all cellular tissues. It is this latter language that generalizes throughout the world of living cellular organisms and that clarifies an important reason why competition occurs, not only to guide natural selection on the level of species, but also to support processing of distributed patterns at every organizational level in all cellular organisms. This latter language is part of a universal developmental code that I will further discuss in Chapter 17.

Weber law, adaptation, and shift property. So far, I have mostly discussed the simplest feedforward, or non-recurrent, shunting on-center off-surround network. Even this network has important properties that are also found in the brain. After describing one of these properties, I will show you how the simplest equation can be altered in easy incremental steps to realize a wide range of additional useful properties. These examples illustrate a kind of conceptual evolution of competitive systems that describes and mechanistically organizes the useful properties of ever-more-complex non-recurrent shunting on-center off-surround networks. I already showed you in Chapter 1 how different feedback signal functions of a feedback, or recurrent, shunting on-center off-surround network (Figure 1.6) can support very different transformations of input patterns into stored short-term memories (Figures 1.7-1.9).

The new property of the simplest non-recurrent network is a manifestation of how solving the noise-saturation dilemma enables it to maintain its sensitivity to inputs even when the off-surround may vary greatly in strength. This *shift property* is most easily seen by plotting each equilibrium activity x_i against the logarithm $K = \ln(I_i)$ of the input I_i to its on-center. The subscript i is not included in the notation K for simplicity. When this is done, $x_i(K, J)$ can be plotted as a function of K for different fixed values $J = \sum_{k \neq i} I_k$ of the total off-surround input. The resulting plots at the top of Figure 2.25 show how each activity x_i increases with increases in K when the total off-surround input J is fixed at successively larger values $J_1, J_2, \ldots$ As J is chosen at successively larger values, the curves of the corresponding activity functions x_i shift to the right; hence the name *shift property*. The main point is that each x_i begins to increase at larger values of K in response to a larger choice of J, but the same maximum activity is eventually approached. This property is also called a Weber law because it shows how the sensitivity of each cell to excitatory on-center inputs is *divided* by the size J of the total inhibitory input to it.

The shift property can be proved with simple high school algebra. The main idea is shown in the bottom equation in Figure 2.25. Here two of the curves in the figure, with different off-surround inputs J_1 and J_2, are equated, one a shift S of the other. Then one solves for S. The resulting shift S is shown at the lower right of Figure 2.25.

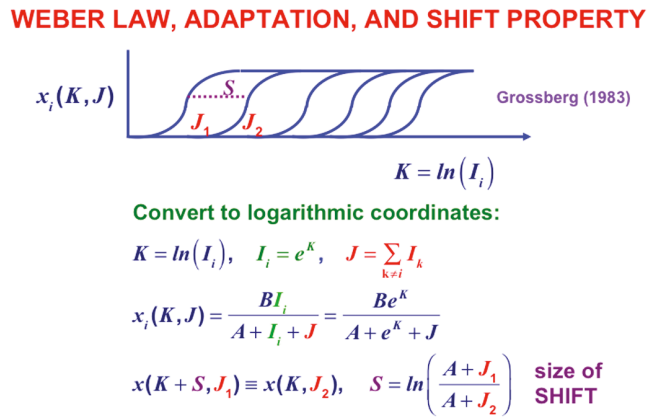

FIGURE 2.25. An on-center off-surround network can respond to increasing on-center excitatory inputs without a loss of sensitivity. Instead, as the off-surround input increases, the region of a cell's maximal sensitivity to an increasing on-center input shifts to a range of larger inputs. This is because the off-surround divides the effect of the on-center input, an effect that is often called a Weber law.

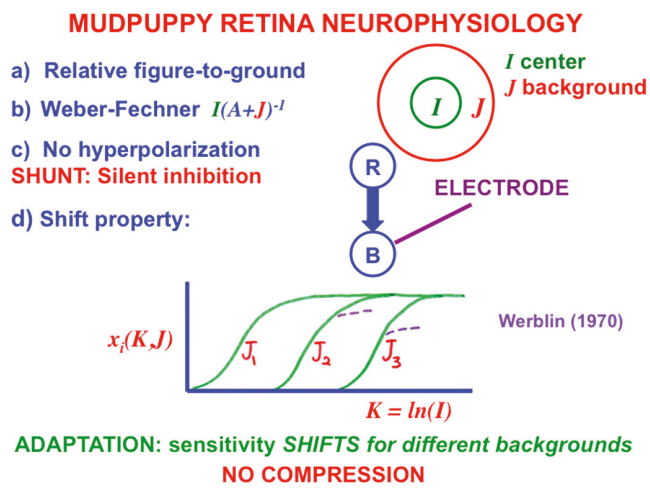

FIGURE 2.26. The mudpuppy retina exhibits the shift property that occurs in the feedforward shunting on-center off-surround network in Figure 2.25. As a result, its sensitivity also shifts in response to different background off-surrounds, and therefore exhibits no compression (dashed purple lines).

Shift property and shunting on-center off-surround network in the mudpuppy retina. In 1971, Frank Werblin published neurophysiological data from the retina of an aquatic salamander called the mudpuppy, or *Necturus maculosus* (Werblin, 1971). His data also describe a shift property when the on-center input is plotted in logarithmic coordinates (Figure 2.26). The main point to emphasize is what did *not* happen: The dynamic range of the cell responses did not get smaller as the off-surround was chosen at larger values, as illustrated by the dashed purple curves in Figure 2.26. Thus, the mudpuppy retina solves the noise-saturation dilemma, leading one to wonder if it does so using a shunting on-center off-surround network. Earlier studies showed that, in fact, an on-center off-surround network does exist in the mudpuppy retina. As illustrated at the bottom of Figure 2.27, retinal photoreceptors (R) excite their target bipolar cells (B) as well as horizontal cells (H) that carry inhibitory off-surround signals to other photoreceptors. Thus, by the time the second

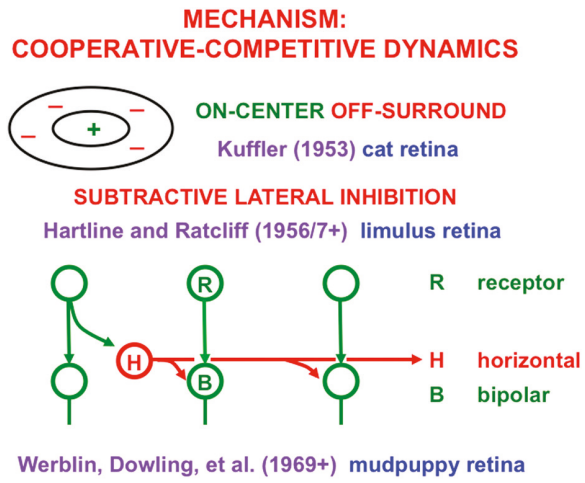

FIGURE 2.27. A schematic of the on-center off-surround network that occurs in the mudpuppy retina, including the three main cell types: Receptors, horizontal cells, and bipolar cells.

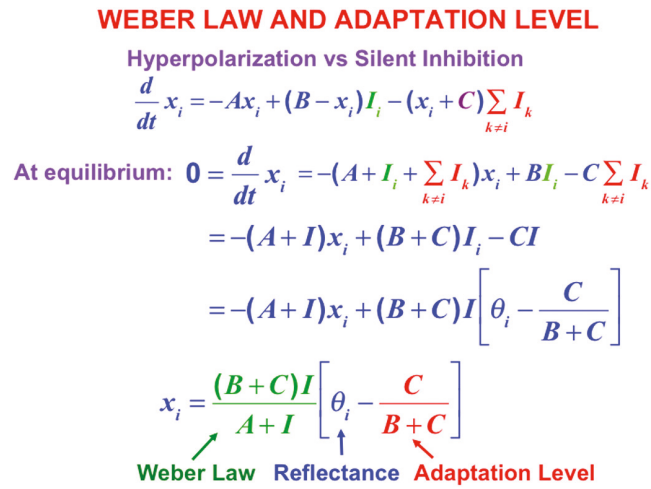

FIGURE 2.28. Silent inhibition is replaced by hyperpolarization when the inhibitory saturating potential is smaller than the passive saturating potential. Then an adaptation level is created that determines how big input ratios need to be to activate their cells.

layer of cells in the retina, the bipolar cells, are activated by light inputs, the noise-saturation dilemma has already been solved.

Intracellular adaptation within photoreceptors creates a Weber law acting in time. The noise-saturation dilemma must actually already be solved in the photoreceptors, even before they activate the bipolar-horizontal cell network. This has to happen because, otherwise, individual photoreceptors could saturate in response to intense inputs even before they could generate outputs to the bipolar cells and horizontal cells. Indeed, we are capable of seeing in bright sunlight as well as in dimly lit rooms, at least after undergoing a suitable interval of adaptation to the new ambient illumination level. Photoreceptor adaptation is, however, *intra*cellular, not *inter*cellular like an on-center off-surround network. Its intracellular process adapts *through time*, rather than *across space* as in the case of an on-center off-surround network, by using a habituative transmitter gate, or medium-term memory (Figures 2.3 and 2.11), to divide current inputs by a time average of previous inputs. In this way, intense ambient illumination causes division of the large current light intensity by a time average of recent large light intensities, thereby normalizing the response via an intracellular Weber law, thereby keeping the cell's responses within its sensitive dynamic range. Thus, both intracellular adaptation within photoreceptors and intercellular adaptation by the on-center off-surround networks that photoreceptors activate both give rise to Weber laws that normalize and maintain cell sensitivity in response to inputs that may vary greatly through time. Interested readers can find a lot of classical neurophysiological data about photoreceptor adaptation and a model that explains them in a 1981 article by Gail Carpenter and myself about light adaptation in turtle cones (Carpenter and Grossberg, 1981).

From silent inhibition to hyperpolarization and adaptation levels. The shift property describes an important property of the simplest shunting on-center off-surround network by using logarithmic coordinates. Additional properties may be realized by refining this network in simple stages. One such refinement replaces the property of *silent inhibition* (Figures 2.23 and 2.24). This property says that the inhibitory saturation voltage V^- and the passive saturation voltage V^p are the same. As a result, no matter how strong off-surround inhibition may become, the activity or potential of a cell can only be driven towards the equilibrium potential that is defined by the passive saturation voltage, but not below it.

Figure 2.28 replaces silent inhibition by choosing the inhibitory saturation voltage V^- to be smaller than the passive saturation voltage V^p, so that activities, or voltages, can be driven by large enough off-surround inhibitory inputs to be smaller than the equilibrium voltage. In the notation of the top equation in Figure 2.28, the choice of the inhibitory saturation voltage C that is greater than zero accomplishes this because activity x_i may then be driven to the negative voltage $-C$ before the term $(x_i + C)$ equals zero, and thus prevents the off-surround from further inhibiting the activity. When activity x_i is less than zero, it is said to be *hyperpolarized*, or less than the equilibrium voltage.

The other equations in Figure 2.28 use the usual method to compute the equilibrium activity that is caused by constant inputs. This is described by the last equation in Figure 2.28. As in the simplest shunting competitive network, a Weber law term $(B + C)I / (A + I)$ multiplies the

relative activity, or ratio, θ_i of inputs. In addition, there is a new term $C/(B+C)$ that is called the *adaptation level*. The activity x_i is positive if θ_i is bigger than the adaptation level, and negative, or hyperpolarized, if it is smaller than the adaptation level.

Adaptation levels require a broken symmetry between excitation and inhibition. This fact immediately raises the question: How does the brain ensure that enough ratios are bigger than the adaptation level? This is necessary to make their activities positive, and some of them large enough both to represent important features and to trigger output signals to other processing levels.

All the ratios are, by definition, less than one. So too is the adaptation level $C/(B+C)$, because both the excitatory saturation voltage B and the inhibitory saturation voltage C are positive. However, if there are n cells in the network, and all the ratios sum to one, then quite a few ratios will have to be less than 1/n for enough of them to be bigger than 1/n.

These considerations imply that the adaptation level $C/(B+C)$ needs to be chosen considerably smaller than one in order for sufficiently many ratios to be positive. As Figure 2.29 summarizes, this implies that B must be much larger than C. In other words, the difference between the smallest possible activity $-C$ and the equilibrium activity 0 is much smaller than the difference between the largest possible activity B and 0, as illustrated in the upper left corner of Figure 2.29. If we consider that an equal distance between these parameters and 0 would be a symmetric solution to how to choose these parameters, then what actually occurs represents a *broken symmetry* between excitatory and inhibition, with excitation favored. Consequences of this broken symmetry between excitation and inhibition are ubiquitous throughout our brains.

Informational noise suppression and the selection of potentially informative activity patterns. One property of great importance can immediately be derived from this broken symmetry. It is one of the properties which ensure that noise will be suppressed before it can contaminate useful information processing. I have already mentioned one kind of noise suppression that is carried out by recurrent shunting on-center off-surround networks. This kind of noise is the familiar kind: random or quasi-random small endogenous activations of neurons that could otherwise interfere with information processing. Figures 1.7-1.9 show how signals that are faster-than-linear at small activity levels can suppress this kind of noise.

There is also another potentially catastrophic kind of noise that can occur even when endogenous or external inputs may be large. I call this kind of noise *informational noise*. Consider the following example of informational noise in order to fix ideas. Suppose that there are multiple inputs converging on individual cells through time. Suppose moreover that, during a given time interval, the total inputs to each of the cells in the network are equal, even though the individual inputs that contribute to these total inputs may differ across cells. This state of affairs is illustrated in the top left graph of Figure 2.30. Such an input pattern is uniform, or has a zero spatial frequency.

A uniform input pattern is uninformative. It does not provide more evidence for some features over others. It would be catastrophic if uniform input patterns could drive new learning across a network, since all featurally-selective LTM traces could be wiped out.

The broken symmetry provides a simple solution to this problem. It can quickly transform a uniform input pattern, no matter how large its shared input size may be, into a zero activity pattern, as illustrated by the figure on the top right of Figure 2.30. Such an activity pattern could not,

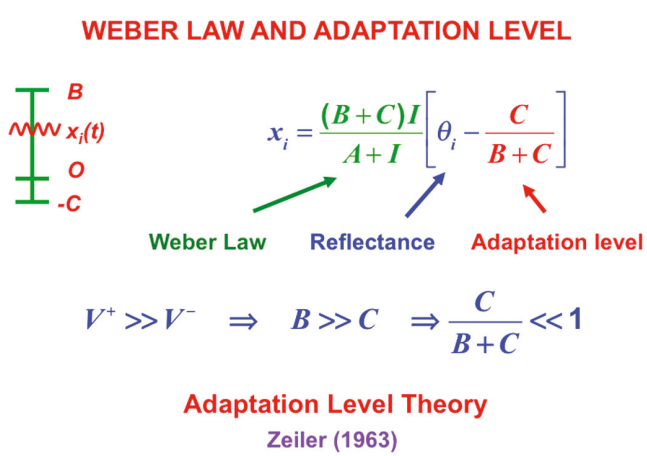

FIGURE 2.29. How the adaptation level is chosen to enable sufficiently distinct inputs to activate their cells.

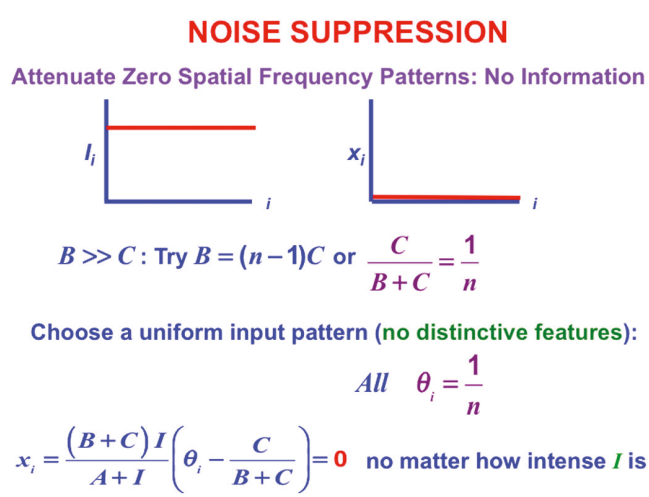

FIGURE 2.30. Choosing the adaptation level to achieve informational noise suppression.

for example, drive any learning via gated steepest descent (Figure 2.11) because all learning gates would be closed. As Chapter 5 will explain, informational noise suppression helps to prevent our brains from spuriously using noise to learn new recognition categories via the instar learning laws that obey the gated steepest descent learning law.

How does the broken symmetry cause informational noise suppression? If the total input to every cell is the same for awhile, then each activity will also become the same, so that all the cell activity ratios will equal 1/n if there are n cells in the network. Noise suppression immediately follows by choosing the adaptation level to also equal l/n (Figure 2.30). Then all activities will equal zero no matter how intense the total input to each cell may be. The choice l/n for the adaptation level will also, in response to a non-uniform input pattern, allow quite a few cell activities to be positive if enough other cells are hyperpolarized.

Informational noise suppression and bottom-up/top-down pattern matching. Informational noise suppression has many useful properties, as well. For example, it suggests how bottom-up and top-down input patterns to a network should be matched. Figure 1.11 called attention to this situation by summarizing some of the dynamics that occur when a learned top-down expectation is sufficiently badly mismatched with a bottom-up input pattern. Chapter 5 will discuss how and why this occurs by introducing and explaining how Adaptive Resonance Theory, or ART, works.

But how should patterns be matched in the first place? The property of informational noise suppression provided an early clue that then evolved into a more precise understanding when ART came on the scene a decade later. Figure 2.31 summarizes the main idea. The top image shows two input patterns that are out of phase with each other, with the most important features of one input pattern occurring where the least important features occur in the other input pattern, and conversely. When these input patterns add at their target cells, the result is a uniform input pattern that is suppressed as informational noise. In other words, input patterns that are out of phase with each other are mismatched and suppressed.

In contrast, if both input patterns are in phase with each other, as in the bottom image of Figure 2.31, then they are added and, in fact, amplified by the automatic gain control that occur during shunting dynamics (Figure 2.32).

These properties were eventually incorporated into the ART Matching Rule that describes how such matching occurs (Figure 1.25). ART went beyond these heuristics by also showing how a top-down expectation, when it acts alone, can modulate, sensitize, or prime target cells, but not fully activate them. Chapter 5 will explain this and much more about how object attention works.

How does informational noise suppression arise during development? Opposites attract! The above examples illustrate how useful the property of information noise suppression is in many brain processes. But how does it arise during development? A big clue, indeed a prediction that goes beyond current data, can be seen by rewriting this property in a different way using the equation in Figure 2.33. This equation is easily derived by setting the adaptation level $C/(B+C)$ equal to 1/n and then rearranging terms.

Rearranging terms is very simple to do, but the implications of the result are potentially profound. This is because the excitatory and inhibitory saturation voltages C and B, respectively, on the left hand side of the equation are *intra*cellular parameters, whereas the numbers *1* and *(n-1)* on the right hand side of the equation are *inter*cellular parameters, indeed the number of on-center (*1*) and off-surround (*n-1*) connections of each input I_i to the network, as illustrated by the figure below the equation.

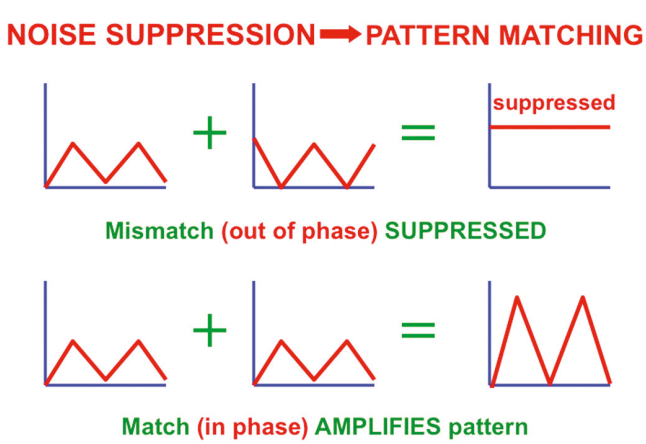

FIGURE 2.31. How noise suppression enables matching of bottom-up and top-down input patterns.

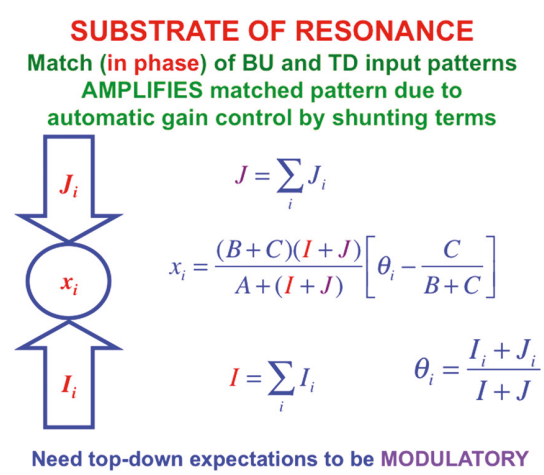

FIGURE 2.32. Matching amplifies the matched pattern due to automatic gain control. See terms *I* and *J* in the equation.

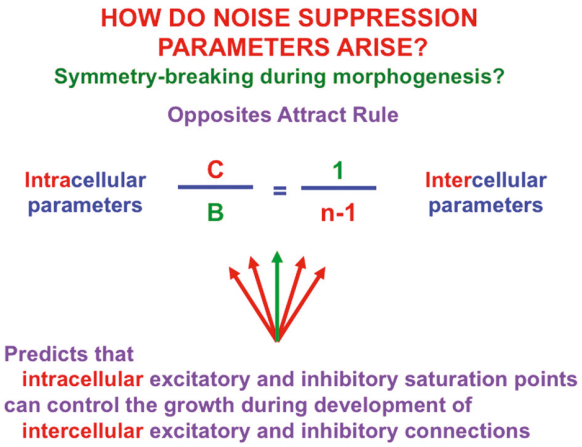

FIGURE 2.33. An opposite-attracts rule during the development of intercellular connections can lead to a mature network that realizes informational noise suppression.

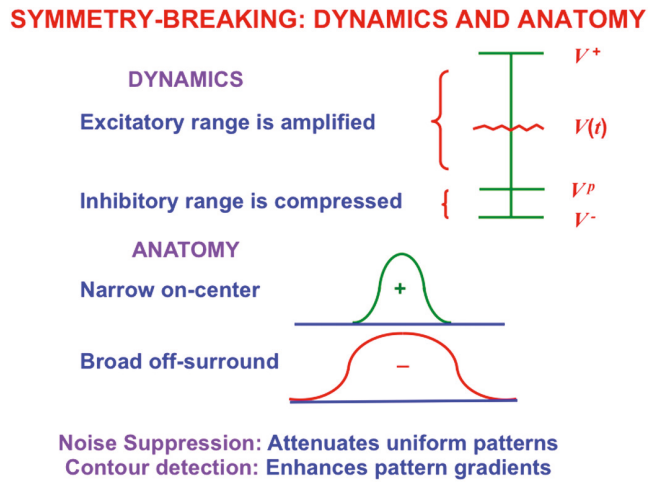

FIGURE 2.34. How to achieve informational noise suppression in a network with multiple parallel processing channels.

How can such an equality emerge as a result of development? The obvious answer is: Use an Opposite Attracts Rule in which the broken *intra*cellular asymmetry between inhibitory and excitatory saturation voltages attracts the growth of excitatory and inhibitory connections until they create and match a broken *inter*cellular asymmetry. Then informational noise suppression will follow automatically.

This way of thinking suggests that varying the degree of asymmetry of depolarization to hyperpolarization in Figure 2.29 can alter how on-center and off-surround connections will grow during development. I do not know any experiments that have tried to explain how these intracellular biophysical parameters and the ensuing network intercellular connectivity are coordinated to achieve informational noise suppression. They would not be easy experiments to do, but the answer should be interesting, especially if deviations from informational noise suppression covary with predicted properties of a cell's adaptation level.

From a global off-surround to a distance-dependent one that preserves noise suppression. The on-center of the network in Figure 2.23 consists of only one cell, or cell population, and the off-surround inhibits all other cells equally. This is a limiting case of the on-center and off-surround connections that are typically found in vivo. A more typical situation is depicted in Figure 2.34. Here, the on-center may be narrow, but nonetheless may include several cells. An input to the ith cell will often excite nearby cells with a connection strength that decreases in a Gaussian manner with distance from the ith cell, as illustrated by the green on-center profile in Figure 2.34. The off-surround is broader than the on-center, but instead of inhibiting all other cells equally, may inhibit all cells around the ith cell, including the ith cell, with a strength that also decreases in a Gaussian manner with distance, but more slowly than the on-center kernel, as illustrated by the red off-surround profile in Figure 2.34.

The connections in Figure 2.34 replace the "single channel" network of Figure 2.23 with a "multiple channel" network in which neighboring cells influence one another, but may be hardly influenced, if at all, by cells that are not within the span of their off-surround. Because the multiple channel anatomy preserves the broken symmetry of narrower on-center than off-surround network connections, it can still interact with the intracellular broken symmetry to achieve informational noise suppression.

The top equation in Figure 2.35 describes such a multiple scale network. Instead of the very narrow on-center and global off-surround that is depicted in the upper left image, there is a Gaussian on-center and a broader Gaussian off-surround in the multiple scale network, as depicted in the upper right image. These excitatory and inhibitory interactions are defined by the Gaussian kernels C_{ki} and E_{ki}, respectively. The equilibrium activity x_i is shown in the lower left of Figure 2.35. In this equation, the total activity I again defined a Weber law effect. Instead of a single ratio θ_i on the right hand side of this equation, as in Figure 2.23, all the ratios θ_k of all the cells V_k, ranging from 1 through n, appear in the numerator of this equation. The ratios θ_k in the numerator are multiplied by kernels F_{ki} that are defined by a weighted *difference* of the on-center and off-surround kernels, which is why it is called a weighted Difference Of Gaussians, or DOG (Figure 2.35). The ratios θ_k in the denominator are multiplied by kernels G_{ki} that are defined by a weighted sum of the on-center and off-surround kernels, which is why it is called a Sum Of Gaussians, or SOG. The SOG normalizes the effect of the DOG. This equation hereby shows how both ratio processing and total activity normalization are realized in a multiple scale network.

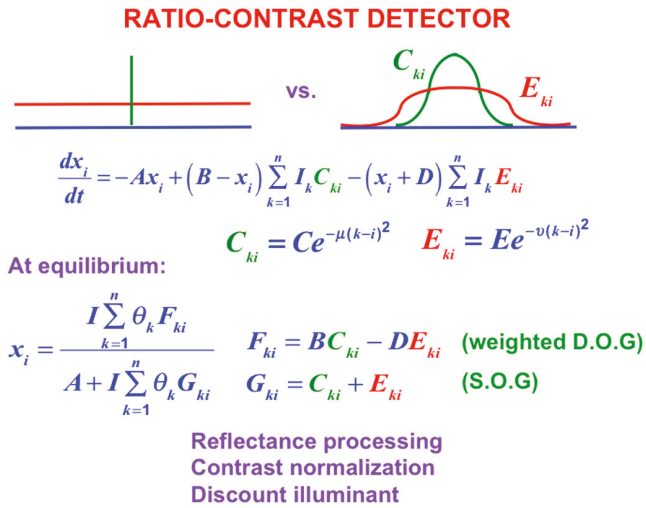

FIGURE 2.35. The equilibrium activities of a shunting network with Gaussian on-center off-surround kernels are sensitive to the ratio-contrasts of the input patterns that they process. The terms in the denominator of the equilibrium activities accomplish this using the shunting on-center and off-surround terms.

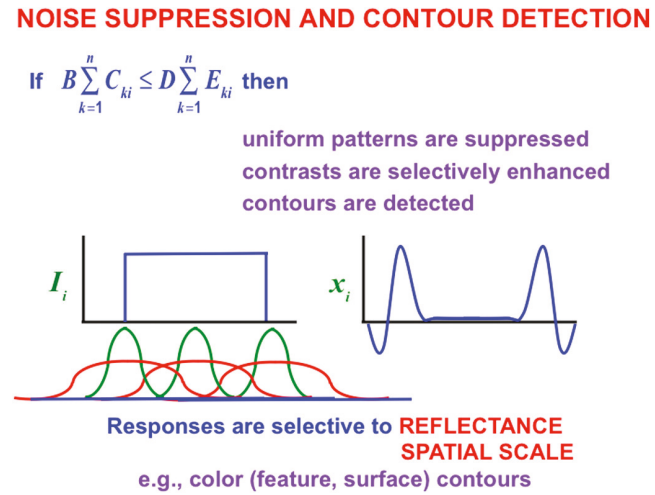

FIGURE 2.36. Informational noise suppression in network with Gaussian on-center and off-surround function as contour detectors that are sensitive to ratio-contrast.

Informational noise suppression implies contour detection in a multiple scale network. Dividing a DOG by a SOG computes the *contrast* of the inputs to each position. In particular, when the network is calibrated to also suppress informational noise, cells selectively respond at positions in the network where input contrasts change sufficiently rapidly across the network. In response to a visual image, for example, such a network responds selectively to positions where there are sufficiently intense luminance edges, textures, or shading. This property will be essential for understanding all of my explanations of how vision works in Chapter 3 and beyond.

Informational noise suppression can easily be realized in a multiple scale network. Simply choose all ratios θ_k to equal 1/n and require that all the resulting equilibrium activities x_i be less than or equal to zero. When this is done, the sums can be carried out separately for the on-center and off-surround kernels in the numerator of the equation for equilibrium activity in Figure 2.35, leading to the inequality $B\sum_{k=1}^{n} C_{ki} \le D\sum_{k=1}^{n} E_{ki}$ at the top left of Figure 2.36. This inequality again shows how intracellular saturation voltages and intercellular connections work together to accomplish informational noise suppression.

When this is done, it is easy to see how this network responds selectively at positions where the input size changes sufficiently rapidly across space, whether due to an edge, texture, or shading gradient. The rectangular input I_i in Figure 2.36 illustrates why this is true. Due to informational noise suppression, cells are not activated at positions either where the input equals zero *or* where the input is large but constant across space. Cells are only activated where the input changes rapidly, as illustrated by the response profile for x_i across position in Figure 2.36.

By including multiple scales, informational noise suppression thus provides us with a contour detector that is sensitive to the relative size of input changes across space. In applications to vision, such a *ratio contrast detector* is sensitive to the reflectances of objects that reflect light to the eyes. This property helps to explain how our brains compensate for variable illumination levels, or "discount the illuminant". I will explain more completely how our brains accomplish this feat in Chapter 4, where I will also show you computer simulations of this and other visual properties that we use every day to accomplish multiple tasks.

How does one evolve a computational brain? My theoretical method

The above discussion illustrates that no single step of theoretical derivation can derive a whole brain. One needs a method for deriving a brain in stages, or cycles, much as evolution has incrementally discovered ever more complex brains over many thousands of years. The following theoretical method has been successfully applied many times since I first used it in 1957. It embodies a kind of conceptual evolutionary process for deriving a brain.

Because "brain evolution needs to achieve behavioral success", we need to start with data that embody indices of behavioral success. That is why, as, as illustrated in Figure 2.37, one starts with Behavioral Data from scores or hundreds of psychological experiments. These data are analyzed as the result of an individual adapting

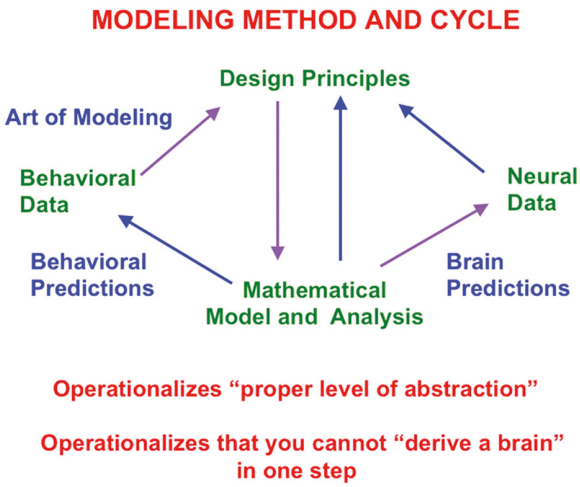

FIGURE 2.37. My models begin with behavioral data, since brains are designed to achieve behavioral success. The text explains how the models evolve in stages, through a process of successive refinements, or unlumpings. These unlumpings together carry out a kind of conceptual evolution, leading to models that can explain and predict ever larger psychological and neurobiological databases.

autonomously in real time to a changing world. This is the Art of Modeling. It requires that one be able to infer from static data curves the dynamical processes that control individual behaviors occurring in real time. One of the hardest things that I teach to my students to do is "how to think in real time" to be able to carry out this speculative leap.

Properly carried out, this analysis leads to the discovery of new Design Principles that are embodied by these behavioral processes. These Design Principles highlight the functional meaning of the data, and clarify how individual behaviors occurring in real time give rise to these static data curves.

These principles are then converted into the simplest Mathematical Model using a *method of minimal anatomies*, which is a form of Occam's Razor, or principle of parsimony. Such a mathematical model embodies the psychological principles using the *simplest* possible differential equations. By "simplest" I mean that, if any part of the derived model is removed, then a significant fraction of the targeted data could no longer be explained. One then analyzes the model mathematically and simulates it on the computer, showing along the way how variations on the minimal anatomy can realize the design principles in different individuals or species.

This analysis has always provided functional explanations and Behavioral Predictions for much larger behavioral data bases than those used to discover the Design Principles. The most remarkable fact is, however, that the behaviorally derived model always looks like part of a brain, thereby explaining a body of challenging Neural Data and making novel Brain Predictions.

The derivation hereby links mind to brain via psychological organizational principles and their mechanistic realization as a mathematically defined neural network. This startling fact is what I first experienced as a college Freshman taking Introductory Psychology, and it changed my life forever.

I conclude from having had this experience scores of times since 1957 that brains look the way that they do because they embody a natural computational realization for controlling autonomous adaptation in real-time to a changing world. Moreover, the Behavior→Principles→Model→Neural derivation predicts new functional roles for both known and unknown brain mechanisms by linking the brain data to how it helps to ensure behavioral success. As I noted above, the power of this method is illustrated by the fact that scores of these predictions about brain and behavior have been supported by experimental data 5-30 years after they were first published.

Having made the link from behavior to brain, one can then "burn the candle from both ends" by pressing both top-down from Behavioral Data and bottom-up from Brain Data to clarify what the model can and cannot explain at its current stage of derivation. No model can explain everything. At each stage of development, the model can cope with certain environmental challenges but not others. An important part of the mathematical and computational analyses is to characterize the *boundary between the known and the unknown*; that is, *which* challenges the model can cope with and which it cannot. The *shape* of this boundary between the known and the unknown helps to direct the theorist's attention to new design principles that have been omitted from the previous analyses.

The next step is to show how these new design principles can be incorporated into the evolving model in a self-consistent way, without undermining its previous mechanisms, thereby leading to a progressively more realistic model, one that can explain and predict ever more behavioral and neural data. In this way, the model undergoes a type of evolutionary development, as it becomes able to cope behaviorally with environmental constraints of ever increasing subtlety and complexity. The Method of Minimal Anatomies may hereby be viewed as way to functionally understand how increasingly demanding combinations of environmental pressures were incorporated into brains during the evolutionary process.

If such an Embedding Principle cannot be carried out—that is, if the model cannot be unlumped or refined in a self-consistent way—then the previous model was, put simply, wrong, and one needs to figure out which parts must be discarded. Such a model is, as it were, an evolutionary dead end. Fortunately, this has not happened to me since I began my work in 1957 because the theoretical method is so conservative. No theoretical addition is made unless it is supported by multiple experiments that cannot be explained in its absence. Where multiple

mechanistic instantiations of some Design Principles were possible, they were all developed in models to better understand their explanatory implications. Not all of these instantiations could survive the pressure of the evolutionary method, but some always could. As a happy result, all earlier models have been capable of incremental refinement and expansion.

The cycle of model evolution has been carried out many times since 1957, leading today to increasing numbers of models that *individually* can explain and predict psychological, neurophysiological, anatomical, biophysical, and even biochemical data. In this specific sense, the classical mind-body problem is now being incrementally solved.

Embedding, unlumping, and correspondence principles

There are several equivalent ways to describe such a refinement procedure: It embodies an *embedding, unlumping,* or *correspondence principle* constraint: One needs to be able to consistently *embed* the previous model into the new expanded model; the previous model needs to be *unlumpable* into an increasingly complex model; a *correspondence principle* characterizes how the variables of a lumped version of the model can be unlumped into the finer variables of a more realistic model.

Correspondence principles are also well known in physics: https://en.wikipedia.org/wiki/Correspondence_principle. One famous one is the correspondence principle that the great Danish physicist, Niels Bohr, used in the early 1900s to link principles of classical mechanics to principles of quantum theory, and then subsequently to quantum mechanics. Another is how Albert Einstein linked principles of classical mechanics to Special Relativity Theory. When a correspondence principle is applied to mind and brain, it places a surprisingly severe test on the adequacy of previously discovered theoretical hypotheses. Many concepts and mechanisms that have been proposed by scientists over the years fail to satisfy the embedding constraint. They seem plausible within a restricted explanatory context, but fall to pieces when they are confronted with more severe environmental challenges. That is why many models come and go with surprising rapidity, and do not get integrated into principled theories of ever greater predictive power. In the past, the main organizational principles I have discovered have survived multiple model unlumping cycles, even while their mathematical realizations in neural circuits may have become increasingly sophisticated and subtle.

I realized when I began my scientific work in 1957 that one would have to derive mind and brain models in successive approximations by using such an embedding constraint. That is why I called my earliest models *Embedding Fields*. The word "embedding" summarized the unlumping constraint. The word "fields" was a short-hand for the fact that interactions within a neural network generate the emergent properties that map onto behavior. These ideas were considered radical in those days, and they took twenty-five years to start to become popular. I wrote a monograph about Embedding Fields during my first year as a graduate student at the Rockefeller Institute in 1964, to which I transferred after studying a lot of graduate mathematics and other scientific topics at Stanford University from 1961-1964 (Grossberg, 1964). This monograph summarized the research that I had done since I was a college Freshman in 1957. It was mailed to 125 of the world's top psychologists and neuroscientists. At that time, the scientific community, by and large, had no idea what to do with it. It can be found at the website http://sites.bu.edu/steveg/files/2016/06/Gro1964EmbeddingFields.pdf.

Many cycles of model evolution have occurred since 1957, with many more colleagues joining this activity starting in the 1980s. All of these evolutionary stages have successfully built a foundation for their progeny. I did not dream when this work began that, in my own lifetime, we could achieve the kind of quantitative understanding of both psychological and neurobiological data that we are experiencing today, including as one example explanatory linkages between fine details of the anatomy and neurophysiology of identified neurons and brain circuits in the cerebral cortex with the psychological functions that they control. Nor did I dream—although I dearly hoped—that this type of work would become so mainstream in the scientific community. The present book will necessarily omit many of these modeling cycles and sociological transformations. I would like you to realize, however, that this process of successive approximations is an essential tool for linking mind to brain, and it still continues today. The remainder of the book will survey some of the results that have been derived with this method, starting with visual perception, since so much of our brain is specialized for seeing. A reader who wants to get a larger sample of the archival technical research articles on which this book is based can scan or download them from my personal web page, sites.bu.edu/steveg.

A never-ending controversy?

I have spent so much time discussing a particular approach to understanding how brains give rise to minds for several reasons. The most direct reason is for the reader to realize that there exists a systematic method for linking brains to minds that has been incrementally developing ever more powerful explanatory and predictive theories of different aspects of biological intelligence for over a half century. I

hope readers of this book will therefore become skeptical about, and challenge, claims that "we know nothing about how the brain works". A no less important reason for discussing this approach is that the question of how to understand mind and brain is still controversial. Productive psychologists and neuroscientists may hold very different opinions about the state of the art and how to advance it. I believe that there are at least two related reasons for this enduring controversy:

First, linking brain to mind is one of the most intellectually challenging endeavors in the history of science. It will take a long time before even simple, and redundantly supported, theoretical explanations of subtle data will be universally appreciated, if only because they are based upon new ways of thinking and neural networks that typically exhibit nonlinear, nonlocal, and nonstationary properties.

Second, despite enormous progress during the last thirty years in building scientific infrastructure to transmit knowledge about our field, including the creation of interdisciplinary educational departments and programs, scientific societies, conferences, and journals, our field still faces a major literacy problem. Part of the reason for this is that theoretical progress has been so fast that it has been difficult to incorporate it into these institutions. A more serious problem is that the deepest theoretical discoveries have been highly interdisciplinary and have unified knowledge about multiple levels of biological intelligence using model neural architectures and their emergent psychological properties. The emergent properties of these modal architectures follow from the dynamics of carefully chosen nonlinear systems of differential equations. Unfortunately, most universities and scientific conferences still do not, by and large, teach the range of interdisciplinary knowledge that is needed to link brain to mind on an advanced level. Even today, although much less than in the past, great experimental neuroscientists may be naïve or dismissive about psychology, and conversely, since that was not part of their own training. And most psychological and neurobiological experimentalists have little or no training in mathematics or how to model. They are simply unqualified to understand, or even to hold an informed opinion about, the state of the art in mind-brain theory. Since a lot of political power in psychology and neuroscience is held by experimentalists, this tends to warp how the field evolves in a way that does not adequately exploit important theoretical discoveries.

These limitations have led people who believe in Big Data to suggest that nothing can be understood theoretically unless we first collect huge amounts of data about every aspect of brain organization. Some neuroscientists also believe that a model is not "realistic" unless it includes every fact from their own laboratories, irrespective of whether these facts are rate-limiting in the explanation and quantitative simulation of important databases about mental functions. These beliefs are contradicted by the cyclic and incremental nature of the modeling Method of Minimal Anatomies depicted in Figure 2.37, and by the huge amount of explanatory and predictive theoretical success of the ensuing models over the past half century.

To overcome these unfounded beliefs, sustained tutorials about theoretical breakthroughs, as is often practiced in theoretical physics, need to be scheduled into major experimental conferences, workshops, and educational programs, and adequately funded to enable theorists, experimentalists, and students of all persuasions to get an opportunity to keep abreast of theoretical advances in our most exciting field as its progress accelerates even more in the years ahead.

From brain theories to technological applications

There is one additional, and very important, feature of the modeling cycle that needs to be described; namely, at *every* stage of the modeling cycle, applications are spun off to be used by engineers and technologists who are interested in developing systems that are capable of increasingly autonomous intelligence and control (Figure 2.38). Why are these mind-brain theories useful to technologists? Every technological application needs both to identify a *function* in the real world that an new application should handle, as well as a *mechanism* with which to realize it. Neural networks that link brain to mind provide both mechanism and function, respectively. Function is derived from the psychological analyses that lead to the discovery of new design principles, whereas mechanism is specified within the neural networks that are derived from these design principles. Moreover, these neural networks are designed to achieve autonomous adaptation in a changing world, and increasing numbers of technological applications require such a competence.

I would argue, moreover, that the development of increasingly autonomous adaptive intelligent agents will become the most important technological revolution of our age, and that insights from biological intelligence will play an increasingly vital role in these developments during the next generation. Finally, unlike many psychological and neurobiological experimentalists, engineers and technologists often have the mathematical and computational training needed to be able to fluently read and understand the mathematical equations that define these neural networks, even if they may not know much about the psychological and neurobiological data that they explain.

Even in technology, however, there are conservative tendencies, as indeed there should be. Engineers usually want to use tried and true methods to solve their problems,

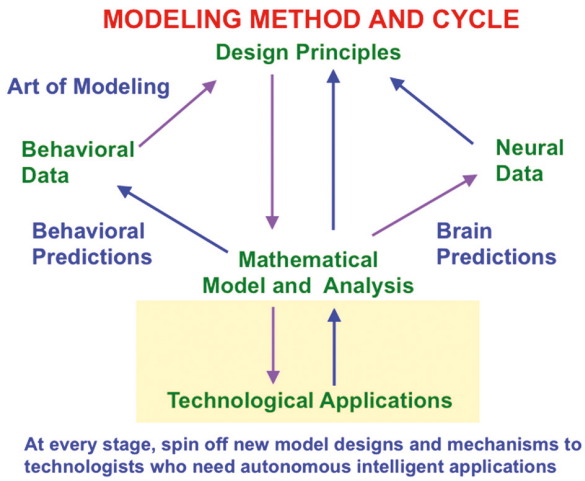

FIGURE 2.38. Our models have been used in many large-scale applications to engineering and technology. Linking brain to behavior explains how brain mechanisms give rise to psychological functions, and do so autonomously. The combination of mechanism, function, and autonomy helps to explain their value in helping to solve outstanding problems in technology.

if only to ensure the reliability of their solutions. They do not want their bridges or airplanes to fall down! However, as increasing numbers of problems arise that are best solved using autonomously intelligent systems that can learn on the fly from huge databases that exhibit rapidly changing statistics and unexpected events, the number of tried and true methods to solve these problems becomes vanishingly small. So new approaches need to be tried and developed. Models that embody the brain's superb autonomously intelligent capabilities are consequently becoming increasingly used. Indeed, most of the government grants that supported my research for over thirty years were interested in discovering new brain designs for autonomous intelligence that could be applied to outstanding technological problems. A lot of scientists, engineers, and companies have used models that Gail Carpenter and I, separately and together, developed with other colleagues with this grant support for both scientific and large-scale applications in technology. A partial list of applications can be found on the CNS Technology website: http://techlab.bu.edu/resources/articles/C5.

It is particularly exciting that some of the biggest technology companies in the world, including Apple and Google, are currently investing heavily in neural networks. Much of that investment is going into trying to solve various Big Data problems by applying Deep Learning. As I noted above, Deep Learning uses the learning algorithm of the earlier back propagation model, which in turn uses a variant of the methods of steepest descent that have been developed by mathematicians since at least the time of Gauss. Thus, the mechanisms that are used in back propagation and Deep Learning are based on classical mathematical ideas. I therefore often call them "neo-classical" models. It should therefore not be a surprise that these models are not consistent with mechanisms that are found in the brain, and that they can experience catastrophic forgetting if they try to learn too quickly, or in a too dense environment, or an environment whose statistics change rapidly and unpredictably. Unfortunately, these properties are often found in real-world Big Data databases. When technology companies fully face the challenges of realizing adaptive autonomous intelligence in rapidly changing environments, my belief is that they will then increasingly turn away from Deep Learning to models like Adaptive Resonance Theory to try to solve their Big Data problems.